

# EECS 841: Computer Vision

## Lecture 8

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# Outline: Part II

## Binary Image Processing

- Morphological Operations
- Labeling Regions
- Computing Distance in a Digital Image
- Region Properties
- Skeletonization
- Boundary Representations



Separating a “common” region for labelling followed by analyzing → ex: lung segmentation in biomedical domain

# Background

- A region have some “connected” components
- So does the boundary
- Thus, it is crucial to realize the connectivity of the pixels
- Connectivity can be expressed by adjacency theory
- Adjacency depends on “neighborhood” theory

# Background: Neighborhood and Adjacency

- The pixels in an image can be represented in terms of the neighbors of a “chosen” pixel

|           |            |          |            |
|-----------|------------|----------|------------|
|           | Col = x-1  | Col = x  | Col = x+1  |
| row = y+1 | (x-1, y+1) | (x, y+1) | (x+1, y+1) |
| row = y   | (x-1, y)   | (x, y)   | (x+1, y)   |
| row = y-1 | (x-1, y-1) | (x, y-1) | (x+1, y-1) |

- If we consider the center element  $(x,y)$ , then every other point can be represented with  $(x,y)$
- This is called neighbor of pixels → representing pixel with their corresponding neighbor
- There are different types of “neighbors”

# Background: 4-Neighbors

- $N_4(P)$  or 4 neighbors: it is the set of  $P$ 's horizontal and vertical neighbors

|           | Col = x-1 | Col = x  | Col = x+1 |
|-----------|-----------|----------|-----------|
| row = y+1 |           | (x, y+1) |           |
| row = y   | (x-1, y)  | (x, y)   | (x+1, y)  |
| row = y-1 |           | (x, y-1) |           |

- $N_D(P)$  diagonal neighbors: it is the set of  $P$ 's diagonal neighbors

|           | Col = x-1  | Col = x | Col = x+1  |
|-----------|------------|---------|------------|
| row = y+1 | (x-1, y+1) |         | (x+1, y+1) |
| row = y   |            | (x, y)  |            |
| row = y-1 | (x-1, y-1) |         | (x+1, y-1) |

Does  $N_D(P)$  also have 4 neighbors?

# Background: Neighborhood and Adjacency

- $N_8(P)$  or 8 neighbors : this is a set where we have union of 4 horizontal-vertical neighbors and the 4 diagonal neighbors.

|           |            |          |            |
|-----------|------------|----------|------------|
|           | Col = x-1  | Col = x  | Col = x+1  |
| row = y+1 | (x-1, y+1) | (x, y+1) | (x+1, y+1) |
| row = y   | (x-1, y)   | (x, y)   | (x+1, y)   |
| row = y-1 | (x-1, y-1) | (x, y-1) | (x+1, y-1) |

- Here we consider all the surrounding pixels of the 'center' pixel
- Thus, :  $N_8(P) = N_4(P) + N_D(P)$

# Connectivity a.k.a Adjacency

- Definition: Two pixels are considered “connected” or “adjacent” if they are neighbors and they have ‘same/similar’ gray level
- There are three major types of adjacent or connectivity: 4 adjacency, 8 adjacency, and mixed adjacency
- **4 adjacency:** if the same/similar pixels in V set are lying in 4 neighborhood (= having  $N_4(P)$ ) of a pixel P → the relationship between P and the other pixel establishing the relationship is known to have “4-adjacency”
- The only elements from V set will be included as “connections”

Question: find a pixel that has the 4-adjacency with P in the following where  $V = \{1\}$

|   |   |       |   |
|---|---|-------|---|
| 0 | 1 | 0     | 1 |
| 0 | 0 | 1 = P | 0 |
| 0 | 0 | 1     | 0 |
| 1 | 0 | 0     | 0 |

|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 0 | 1 |
| 0 | 0 | 1 | 0 |
| 0 | 0 | 1 | 0 |
| 1 | 0 | 0 | 0 |



= this connection is called 4 adjacency

# Connectivity aka Adjacency

- 8 adjacency: if the same/similar pixels in V set are lying in the 8 neighborhood (= having NP(8)) of P then they are having 8-adjacency connection → only elements from V set will be included as “connections”

Question: find a pixel that has the 8-adjacency with P in the following where  $V = \{1\}$

|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 0 | 1 |
| 0 | 0 | 1 | 0 |
| 0 | 0 | 1 | 0 |
| 1 | 0 | 0 | 0 |

|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 0 | 1 |
| 0 | 0 | 1 | 0 |
| 0 | 0 | 1 | 0 |
| 1 | 0 | 0 | 0 |



= this connection is called 8 adjacency



# Practice

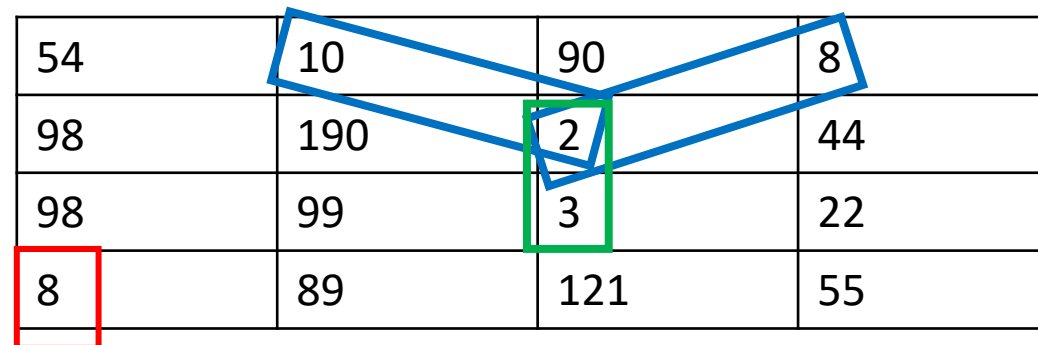
Question: Show all possible adjacencies for the following image for  $V = \{1:10\}$

or,

i) find all the 4 adjacency in the following where  $V = \{1:10\}$

ii) find all the 8 adjacency in the following where  $V = \{1:10\}$

- The connections can be made only with the consideration of the elements under V set



|    |     |     |    |
|----|-----|-----|----|
| 54 | 10  | 90  | 8  |
| 98 | 190 | 2   | 44 |
| 98 | 99  | 3   | 22 |
| 8  | 89  | 121 | 55 |

This element is in set V but it does not have any adjacent neighbors, thus it is “Not adjacent”

# Mixed adjacency (m-adjacency)

- One of the goals of 4 and 8 adjacency was to see how many connections are there for P
- Now there is another theory “mixed adjacency” which deals with a scenario where multiple connections can be possible between the elements of set V but we choose a unique one which connects all possible ones and in shortest path possibly (considering 4 adjacency first and then 8 adjacency)
- Question:  
only using “1”s how many connections can you make?

|   |   |   |
|---|---|---|
| 0 | 1 | 1 |
| 0 | 1 | 0 |
| 0 | 0 | 1 |

# Mixed adjacency (m-adjacency)

- One of the goals of 4 and 8 adjacency was to see how many connections are there for P
- Now there is another theory “mixed adjacency” which deals with a scenario where multiple connections can be possible between the elements of set V but we choose a unique one which connects all possible ones and in shortest path possibly (considering 4 adjacency first and then 8 adjacency)

- examples

only using “1”s we are making the connections

|   |   |   |
|---|---|---|
| 0 | 1 | 1 |
| 0 | 1 | 0 |
| 0 | 0 | 1 |

|   |   |   |
|---|---|---|
| 0 | 1 | 1 |
| 0 | 1 | 0 |
| 0 | 0 | 1 |

|   |   |   |
|---|---|---|
| 0 | 1 | 1 |
| 0 | 1 | 0 |
| 0 | 0 | 1 |

# Mixed adjacency (m-adjacency)

- There are multiple ways to connect the elements.  
If two pixel can have both 4 and 8 adjacency, prioritize 4 adjacency to connect them.  
Keep connecting the pixels following the rule.
- Finally, you find a unique **path** that connects the elements.
- Question: Connect the pixels using “1”s.  
only using “1”s we are making the corrections

|   |   |   |
|---|---|---|
| 0 | 1 | 1 |
| 0 | 1 | 0 |
| 0 | 0 | 1 |

Ans:

|   |   |   |
|---|---|---|
| 0 | 1 | 1 |
| 0 | 1 | 0 |
| 0 | 0 | 1 |

# Distance Measures

|          |  |  |          |
|----------|--|--|----------|
|          |  |  | $q(s,t)$ |
|          |  |  |          |
|          |  |  |          |
| $p(x,y)$ |  |  |          |

- We have two points  $p$  and  $q$  with coordinates  $(x,y)$  and  $(s,t)$  in the image, can we measure distance
- There are multiple ways to measure distance between pixels.
- The 4 popular ones are:
  - Euclidean distance
  - City block
  - Chessboard
  - M-distance
- Among them, the first 3 are independent of “path adjacency presence”
- The last one counts on “m-adjacency”

# General rules for Distance Measurement

Path: if the pixels under  $p$  and  $q$  are in  $V$  set

You can ignore the path determination between  $p$  and  $q$  while finding the city block, chessboard and Euclidean

Thus, they are independent of path related information (e.g., consideration of  $V$  set)

But for  $m$ -distance the points have to be in  $V$  and connected through the members, that's why path consideration is important.

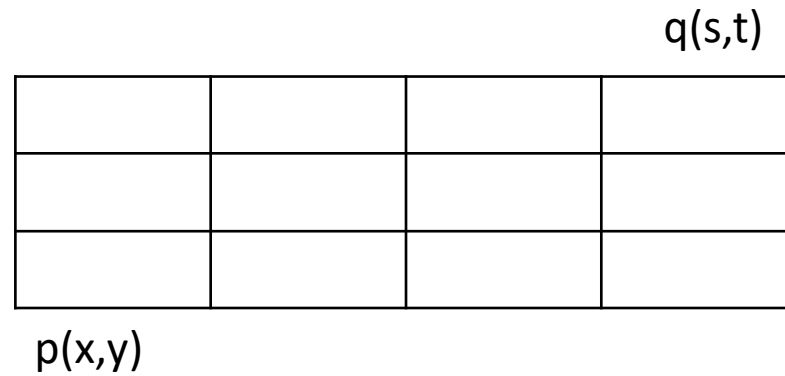
# Distance Measures: Euclidean Distance

|        |  |  |  |        |
|--------|--|--|--|--------|
|        |  |  |  | q(s,t) |
|        |  |  |  |        |
|        |  |  |  |        |
|        |  |  |  |        |
| p(x,y) |  |  |  |        |

$$D_E(p, q) = \sqrt{(x - s)^2 + (y - t)^2}$$

”

# Distance Measures: City block

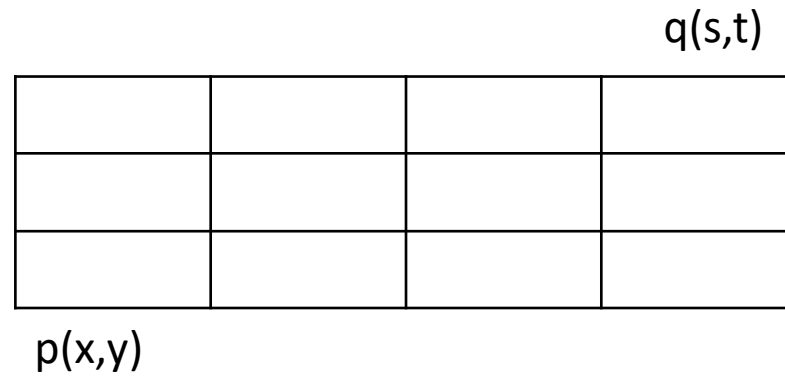


$$D_4(p, q) = |x - s| + |y - t|$$

This is also known as D4 distance, because it measures the distance as if it was trying to connect the two pixels  $p$  and  $q$  following a D4 method and considering unit distance between them.



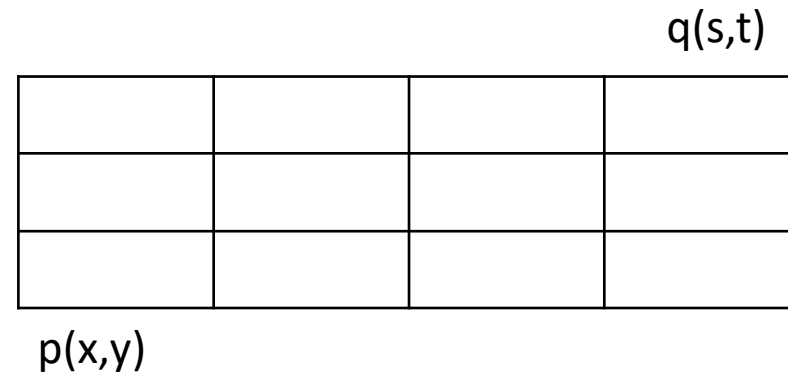
# Distance Measures: Chessboard



$$D_8(p, q) = \max(|x - s|, |y - t|)$$

This is also known as D8 distance, because it measures the distance as if it was trying to connect the two pixels  $p$  and  $q$  following a D8 method and considering unit distance between each connection.

# M-distance



- The distance is known as mixed distance and it is the distance considering the shortest m path between 2 points (p and q)
- The distance depends on the values of the pixels neighborhood pixels → because we need to make sure that the values are part of  $V$  to determine if they really make a path (=m-adjacency)

# General rules for Distance Measurement

You can ignore the path determination between  $p$  and  $q$  while finding the city block, chessboard and Euclidean

Thus, they are independent of path related information (e.g., consideration of  $V$  set)

Path: if the pixels under  $p$  and  $q$  are in  $V$  set

But for  $m$ -distance the points have to be in  $V$  and connected through the members, that's why path consideration is important.

# Practice:

Consider the image segment. If  $V=\{2,3\}$  is the set of gray level values defining the connectivity in the image, then compute D4, D8 and Dm distance between p and q.

Coordinates of p(x,y) = (0,0)

Coordinates of q(s,t) = (4,4)

|       |   |   |   |       |
|-------|---|---|---|-------|
| 2 (p) | 3 | 2 | 6 | 9     |
| 6     | 2 | 3 | 6 | 2     |
| 5     | 3 | 3 | 0 | 5     |
| 2     | 4 | 3 | 7 | 2     |
| 5     | 5 | 2 | 3 | 6 (q) |

Sol:

$$D4(p,q) = |x-s| + |y-t| = |0-4| + |0-4| = 4+4=8 \text{ units.}$$

$$D8(p,q) = \max(|x-s|, |y-t|) = \max(4,4) = 4 \text{ units}$$

There is no connection between p and q hence Dm can't be calculated as q is not included in V  $\rightarrow$  thus, can't compute Dm

# Practice:

Consider the image segment. If  $V=\{0,1\}$  is the set of gray level values defining the connectivity in the image, then compute D4, D8 and Dm distance between p and q.

|       |   |   |       |
|-------|---|---|-------|
| 5     | 1 | 2 | 1 (q) |
| 0     | 2 | 0 | 2     |
| 1     | 2 | 1 | 1     |
| 1 (p) | 0 | 1 | 2     |

Coordinates of  $p(x,y) = (0,0)$

Coordinates of  $q(s,t) = (3,3)$